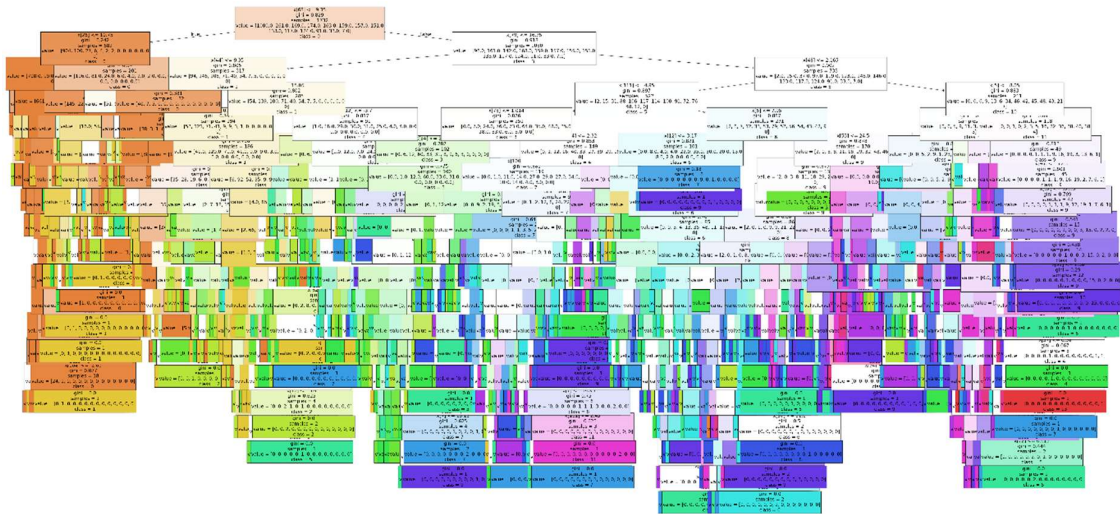


Task 2.4

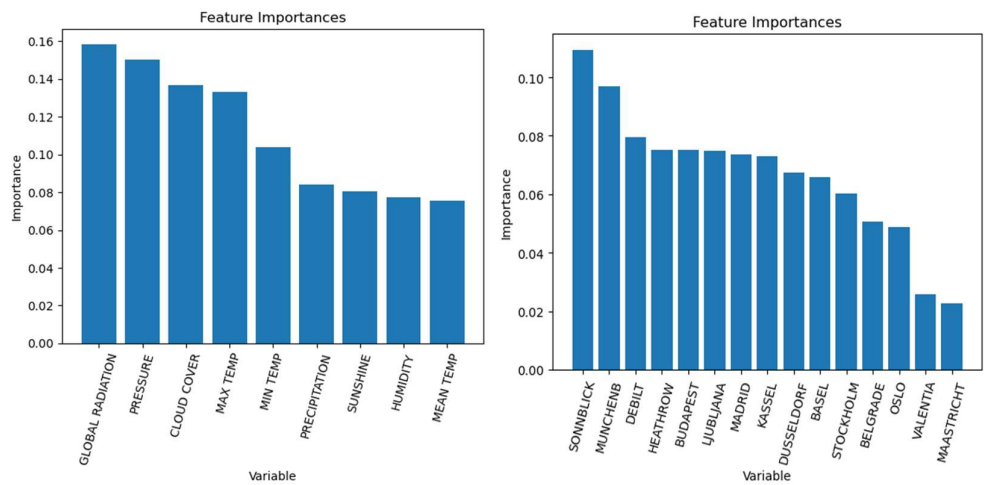
Optimizing Parameters for the Random Forest Model using Pleasant Weather Data

All Weather Stations, 2010-2020

Before Optimization

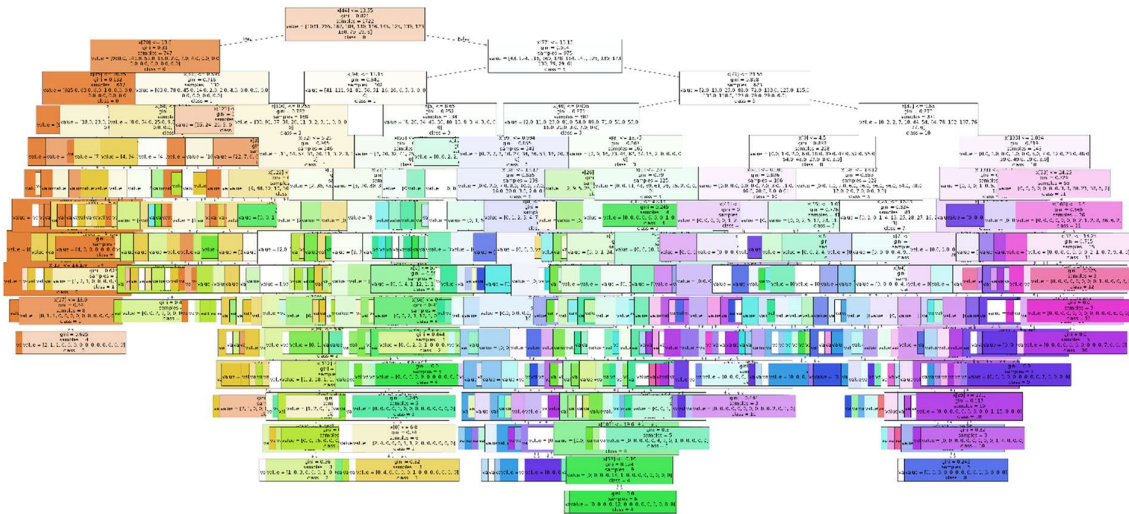


The model showed 55.1% accuracy.

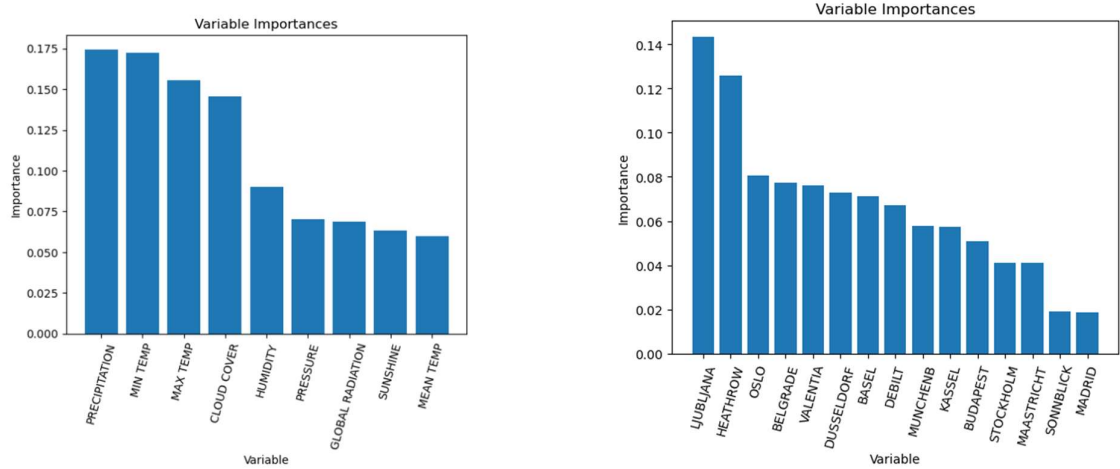


Global Radiation and Pressure were the most important features for this classifier model, followed by Cloud Cover and Maximum Temperature. Among weather stations, Sonnblick had the most influence on the results of the classifier, followed by München and Deilt.

Hyperparameters Determined by Grid Search Optimizer



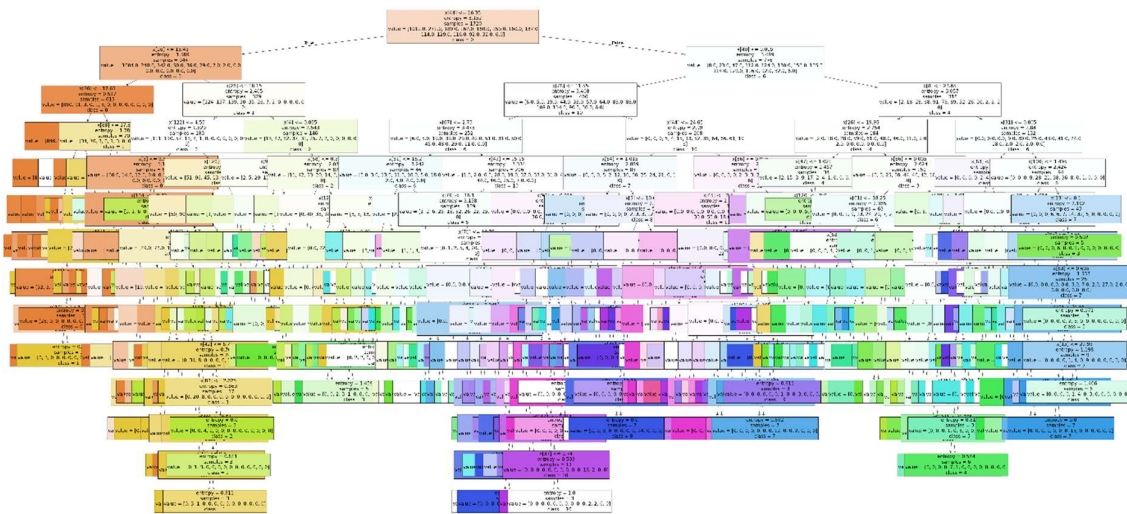
With the best hyperparameters as determined by the grid search optimizer, this model has a value of 57.4%



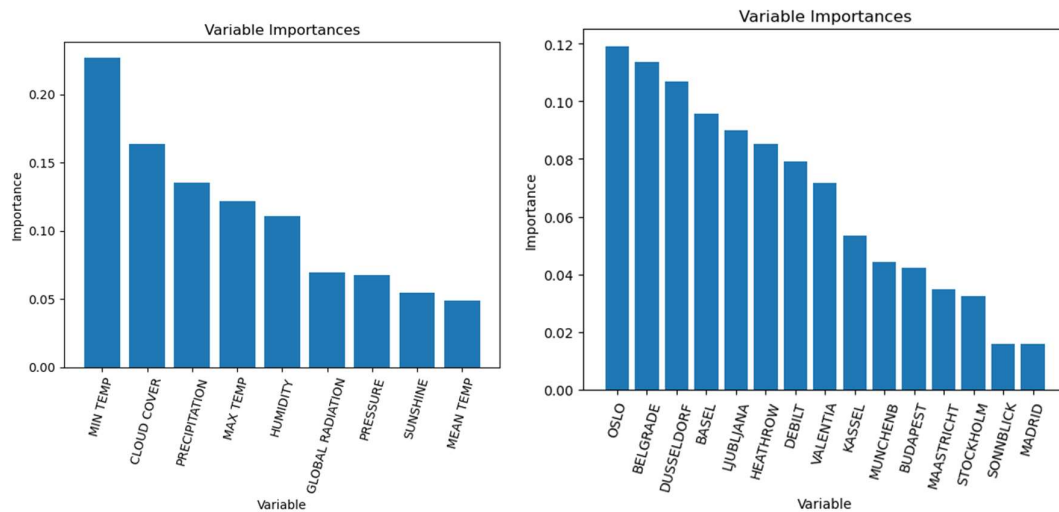
The feature importances for this model show that, out of the weather conditions, precipitation and minimum temperature had the most influence on the classifier.

Among weather stations, Ljubljana and Heathrow had the most significant effect on the classifier.

Hyperparameters Determined by Random Search Optimizer



The random forest model showed 58.1% accuracy with these parameters.



Minimum temperature has a much more significant effect on the classifier compared to other weather conditions. Cloud cover, precipitation and maximum temperature still have a significant influence, though not as much as with the parameters determined by the grid search.

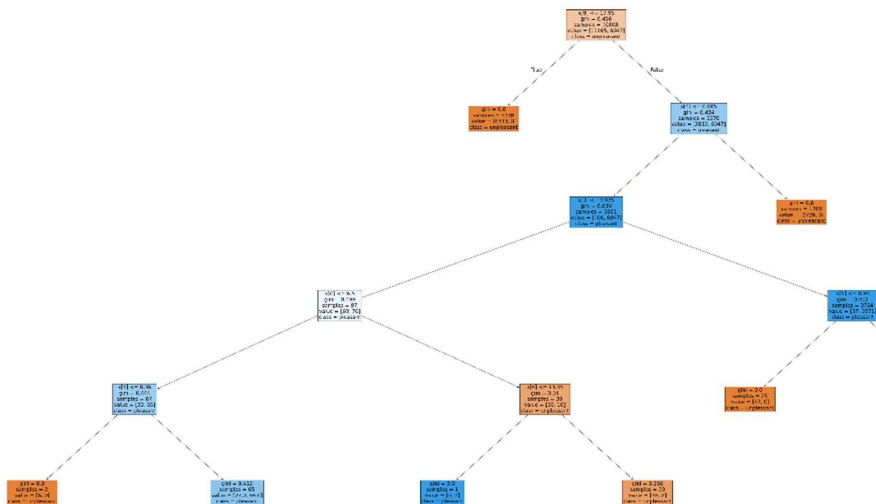
Oslo and Belgrade have the most influence out of all weather stations, whereas with the grid search-determined best hyperparameters, they were the 3rd and 4th most important features. The distribution of feature importance appears to be more uniform compared to that of the model with the previous set of hyperparameters.

The least important stations appear to be mostly the same for both models.

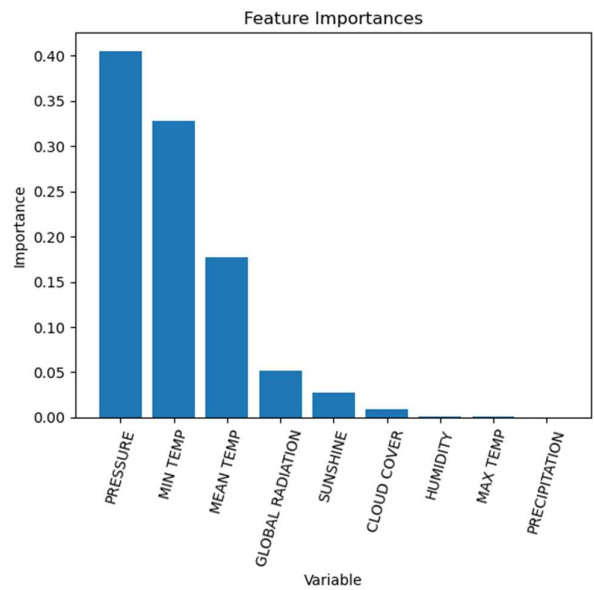
Belgrade, 1960-2022

I chose Belgrade, as this station ranked highly in feature importance for the 2010-2020 data before and after optimization.

Random Forest with Best Hyperparameters according to Grid Search Optimizer

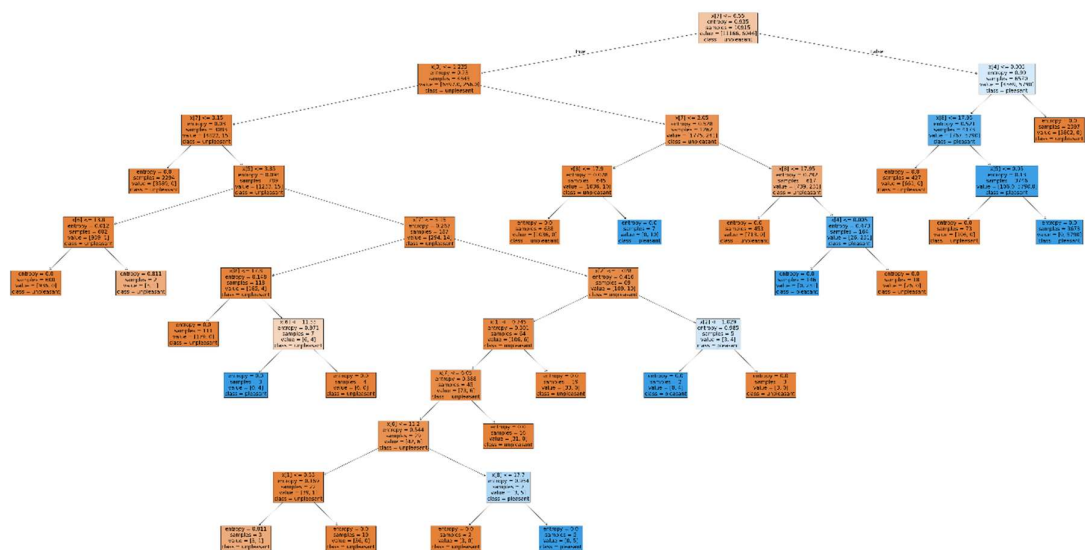


This model showed 100% accuracy.

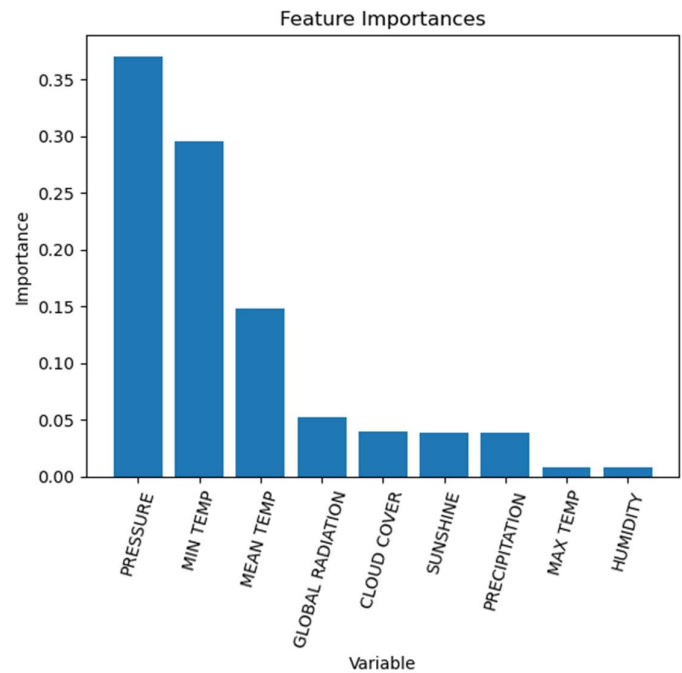


With the hyperparameters set according to the grid search optimizer, pressure and minimum temperature show a significantly high level of influence over the results of the classifier, while humidity, maximum temperature and precipitation appear to have little to no effect at all.

Random Forest with Best Hyperparameters according to Random Search Optimizer



This model showed 100% accuracy again. The tree appears significantly more complex, with many more branches and leaves.



The most important features remained the same as with the grid-search-optimized hyperparameters. However, the least important features did increase somewhat in importance.

Note on Feature Importances

I noticed that, regardless of parameter optimization, the feature importances for this data were highly unstable, with the rankings changing every time I ran the model, even if the accuracy itself remained constant. Adding `random_state` parameters to the functions did not help, nor did setting `bootstrap` to `False`. This leads me to believe that the data itself may be causing this instability, making it difficult to determine any weather conditions or location data that would have a particular influence on the random forest's classifications.

One possible reason for this may be the fact that the Pleasant Weather data is based on local surveys, and that the answers may not be as much influenced by the weather conditions themselves as by other factors, such as cultural attitudes or socioeconomic wellbeing.

Convolutional Neural Network (CNN) with Bayesian-optimized hyperparameters

Hyperparameters Before Optimization:

```
epochs = 60
```

```
batch_size = 9
```

```
n_hidden = 64
```

```
model = Sequential()
```

```
model.add(Conv1D(n_hidden, kernel_size=3, activation='relu', input_shape=(timesteps, input_dim)))
```

```
model.add(Dense(32, activation='relu'))
```

```
model.add(MaxPooling1D())
```

```
model.add(Flatten())
```

```
model.add(Dense(n_classes, activation='relu'))
```

```

180/180 0s 2ms/step
Pred      BASEL  BELGRADE  BUDAPEST  DEBILT  HEATHROW  KASSEL  LJUBLJANA  \
True
BASEL      3      1216      309      132      4      3      1431
BELGRADE   0       783       31       0       0       0       276
BUDAPEST   0       141        9       0       0       0        64
DEBILT      0       55        3       0       0       0        23
DUSSELDORF 0       12        1       0       0       0        16
HEATHROW    0       26        2       1       0       0        49
KASSEL      0        8        1       0       0       0         2
LJUBLJANA   0       28        2       0       0       0        31
MAASTRICHT  0        2        0       0       0       0         6
MADRID      0      126       13      13       0       0      253
MUNCHENB    0        5        1       0       0       0         2
OSLO        0        3        0       0       0       0         1
STOCKHOLM   0        3        0       0       0       0         1
VALENTIA    0        0        0       0       0       0         0

Pred      MAASTRICHT  MADRID  MUNCHENB  OSLO  SONNBLICK  STOCKHOLM  VALENTIA
True
BASEL      59      294        4      205        10         1        11
BELGRADE   0        2        0       0         0         0         0
BUDAPEST   0        0        0       0         0         0         0
DEBILT      0        1        0       0         0         0         0
DUSSELDORF 0        0        0       0         0         0         0
HEATHROW    1        3        0       0         0         0         0
KASSEL      0        0        0       0         0         0         0
LJUBLJANA   0        0        0       0         0         0         0
MAASTRICHT  0        1        0       0         0         0         0
MADRID      5       48        0       0         0         0         0
MUNCHENB    0        0        0       0         0         0         0
OSLO        0        1        0       0         0         0         0
STOCKHOLM   0        0        0       0         0         0         0
VALENTIA    1        0        0       0         0         0         0

```

Without optimized hyperparameters, this model predicted 14 out of 15 stations with very low accuracy (12.7%) and a very high level of loss (16946712576).

After running the data through a Bayesian Optimizer, I found the following optimal hyperparameter values:

activation: 'tanh',

batch_size: 828,

dropout: 0.19967378215835974,

dropout_rate: 0.15427033152408348,

epochs: 67,

kernel: 1.0929008254399954,

layers1: 2,

layers2: 1,

learning_rate: 0.07440107705542671,

neurons: 95,

normalization: 0.9656320330745594,

optimizer: 'nadam'

Pred True	BASEL	BELGRADE	BUDAPEST	DEBILT	DUSSELDORF	HEATHROW	KASSEL	\
BASEL	3561	66	6	10	7	7	0	
BELGRADE	120	944	7	4	1	2	0	
BUDAPEST	19	23	146	12	1	5	0	
DEBILT	9	5	7	56	2	1	0	
DUSSELDORF	4	2	1	2	5	13	0	
HEATHROW	8	2	1	1	5	55	0	
KASSEL	5	2	1	0	3	0	0	
LJUBLJANA	12	2	4	0	0	2	1	
MAASTRICHT	2	0	0	0	0	1	0	
MADRID	38	11	8	0	0	5	2	
MUNCHENB	5	1	0	0	0	0	0	
OSLO	3	0	0	0	0	0	0	
STOCKHOLM	2	0	0	0	0	0	0	
VALENTIA	1	0	0	0	0	0	0	

Pred True	LJUBLJANA	MADRID	MUNCHENB	OSLO
BASEL	1	22	1	1
BELGRADE	3	11	0	0
BUDAPEST	1	6	1	0
DEBILT	1	1	0	0
DUSSELDORF	0	2	0	0
HEATHROW	1	8	0	1
KASSEL	0	0	0	0
LJUBLJANA	26	13	0	1
MAASTRICHT	0	6	0	0
MADRID	5	389	0	0
MUNCHENB	0	0	2	0
OSLO	0	2	0	0
STOCKHOLM	0	0	1	1
VALENTIA	0	0	0	0

The model with these hyperparameters performed with 93.5% accuracy with a loss of 0.191, yet only recognized 11 out of 15 stations.

Iteration

The Air Ambulance company would benefit from the European climate data being broken down into smaller, easily manageable portions when predicting optimal flying conditions. One way to do this would be to group the locations by regional climate type. For example, locations like Budapest, Ljubljana and Belgrade are all located inland, in southeastern Europe, and are therefore likely to experience a humid, continental climate with cold, snowy winters, hot summers and distinct seasons. Locations like Sonnblick and Basel are close to the Alps and would therefore be subject to weather conditions caused by proximity to mountain ranges, such as sudden precipitation, high levels of sunshine and cooler mean temperatures.

For classification of weather data into optimal or sub-optimal flying conditions, a random forest model with optimized parameters seems to be the best option, as it is easier to interpret, less prone to overfitting and can handle a complex set of data. A convolutional neural-network may be more suited to larger sets of data such as the full range of European climate data across all decades. However, this type of model is more complicated to set up and optimize, and often does not recognize the full range of features in the data.

The most important variables for the Air Ambulance company to consider when determining safe flying conditions would be wind-speed (high winds can affect the stability of aircraft), cloud cover, humidity and precipitation (foggy and rainy conditions can obscure a pilot's vision).